MakeNoiseThreshold<MapDomain<AtomDomain<TK>, AtomDomain<IBig>>, MI, MO> for RV

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of MakeNoiseThreshold for RV over hashmaps of big integers in mod.rs at commit f5bb719 (outdated[1]).

This is the core implementation of all variations of the thresholded gaussian or laplace mechanism.

1 Hoare Triple

Precondition

Compiler-verified

MakeNoise is parameterized as follows:

- DI is MapDomain<AtomDomain<TK>, AtomDomain<IBig>>
- MI is LOPInfDistance<P, AbsoluteDistance<UBig>>
- MO is MO

The following trait bounds are required:

- Generic TK implements trait Hashable
- Const-generic P is of type usize
- Generic MO implements trait Measure
- Type ZExpFamily<P> implements trait NoiseThresholdPrivacyMap<LOPInfDistance<P, AbsoluteDistance<UBig>>, MO>

Caller-verified

None

[1] See new changes with `git diff f5bb719..c49be303 rust/src/measurements/noise_threshold/mod.rs`

Pseudocode

```
# analogous to impl MakeNoise<VectorDomain<AtomDomain<IBig>>, MI, MO> for RV in Rust
class RV:
    def make_noise_threshold(
        self,
        input_space: tuple[MapDomain[AtomDomain[TK], AtomDomain[IBig]], MI],
        threshold: IBig,
    ) -> Measurement[
        MapDomain[AtomDomain[TK], AtomDomain[IBig]], HashMap[TK, IBig], MI, MO
    ]:
        input_domain, input_metric = input_space
        output_measure = MO.default()
        threshold_magnitude = threshold.into_parts()[1]  #
        privacy_map = self.noise_threshold_privacy_map(  #
            input_metric, output_measure, threshold_magnitude
        )

        match threshold.sign():
            case Sign.Positive:
                keep_ordering = Ordering.Greater
            case Sign.Negative:
                keep_ordering = Ordering.Less

        def function(data: HashMap[TK, IBig]) -> HashMap[TK, IBig]:
            out = []
            for k, v in data.items():
                if v == 0:  #
                    continue
                v = self.sample(v)  #

                if v.cmp(threshold) == keep_ordering:
                    out.append((k, v))
            # shuffle the output to avoid leaking the order of the input
            random.shuffle(out)  #
            return dict(out)

        return Measurement.new(
            input_domain,
            Function.new_fallible(function),
            input_metric,
            output_measure,
            privacy_map,
        )
```

Postcondition

Theorem 1.1. For every setting of the input parameters (`self, input_space, threshold, MO, TK, P`) to `make_noise_threshold` such that the given preconditions hold, `make_noise_threshold` raises an error (at compile time or run time) or returns a valid measurement. A valid measurement has the following properties:

1. (Data-independent runtime errors). For every pair of members x and x' in `input_domain`, `invoke`(x) and `invoke`(x') either both return the same error or neither return an error.

2. (Privacy guarantee). For every pair of members x and x' in `input_domain` and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for
`output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map`(`d_in`) does not raise an error, and `privacy_map`(`d_in`) $=$ `d_out`, then `function`(x), `function`(x') are `d_out`-close under `output_measure`.

Proof of data-independent errors. The precondition of `Sample.sample` requires that `self` is a valid distribution. This is satisfied by the postcondition of `NoiseThresholdPrivacyMap<MI, MO>` on line 13. The postcondition of `Sample.sample` guarantees that the function only ever returns an error independently of the data. □

For the proof of the privacy guarantee, start by reviewing the postcondition of `NoiseThresholdPrivacyMap<MI, MO>`, which has an associated function `noise_threshold_privacy_map` called on line 13.

Lemma 1.2 (Postcondition of `NoiseThresholdPrivacyMap`)**.** Given a distribution `self`, returns `Err(e)` if `self` is not a valid distribution. Otherwise the output is `Ok(privacy_map)` where `privacy_map` observes the following:

Define `function(x)` as a function that discards all pairs with $v_i = 0$, then updates each remaining pair $(k_i, v_i + Z_i)$, where Z_i are iid samples from `self`, and retains pairs only when $v_i + Z_i$ is strictly beyond `threshold` in its signed direction. The ordering of returned pairs is independent from the input ordering.

For every pair of elements x, x' in `MapDomain<AtomDomain<TK>, AtomDomain<IBig>>`, and for every pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_measure`, if x, x' are `d_in`-close under `input_metric`, `privacy_map(d_in)` does not raise an exception, and `privacy_map(d_in)` $\leq$ `d_out`, then `function(x)`, `function(x')` are `d_out`-close under `output_measure`.

Proof of privacy guarantee. Assuming line 13 does not fail, then the returned privacy map is subject to Theorem 1.2. The privacy guarantee applies when the pseudocode matches the algorithm specification, where Z_i are iid samples from `self`. In this case `self` describes the noise distribution.

We argue that `function` is consistent with the function described in Lemma 1.2. Line 26 omits explicit zero-valued entries before sampling. Under `L0PInfDistance`, missing keys are interpreted as zero, so omitting zero-valued entries preserves all pairwise input distances. Line 28 calls `self.sample(v)` only on the remaining values. The precondition that `self` represents a valid distribution is satisfied by the postcondition of Lemma 1.2; the distribution is valid when the construction of the privacy map does not raise an exception. Since the preconditions for `Sample.sample` are satisfied, the postcondition claims that either returns an error independently of the input `v`, or `v + Z` where `Z` is a sample from the distribution defined by `self`. The keys are then shuffled on line 33 to ensure that the output is independent of the input ordering. In the Rust implementation, a random hasher is used to ensure that the output ordering is independent of the input ordering. This is consistent with Lemma 1.2.

Therefore, the privacy guarantee from Lemma 1.2 applies to the returned measurement. □